

1. EXECUTIVE SUMMARY & ABSTRACT

Edu Growth is an advanced academic analytics platform designed to transition educational monitoring from *reactive* (post-exam grading) to *proactive, predictive intervention*. Traditional learning management systems rely on lagging indicators, making it virtually impossible to assist struggling students during critical learning windows. Edu Growth addresses this systemic gap by continuously evaluating multi-dimensional behavioral, attendance, and granular academic data across **10 Core Data Ingestion Pillars** powered by **15 deep analytical engines**.

The platform delivers role-based, actionable intelligence through three dedicated interfaces: Student Panel, Teacher Panel, and Head of Department (HOD) Panel, enabling synchronized academic support across institutional tiers.

2. SYSTEMATIC DATA INGESTION ENGINE (10 CORE PILLARS)

Edu Growth normalizes and processes multi-source academic signals to construct a holistic 360-degree performance vector for every student:

1. Unit-Wise Marks Granular academic assessment per chapter/unit to pinpoint specific topic degradation rather than coarse course-level failures.	2. Attendance Trends Real-time tracking of physical and virtual classroom presence, capturing subtle decline patterns before critical thresholds.
3. Cumulative CGPA Longitudinal historical academic baseline, establishing individual achievement trajectories and expectations.	4. Overall Class Performance Cohort-level statistical benchmarking (mean, variance, percentile distributions) to contextualize individual effort.
5. Practical / Lab Performance Hands-on application and execution metrics, evaluating practical proficiency alongside theoretical comprehension.	6. Assignment Behavior Submission timeliness, turn-in consistency, and delay ratios serving as leading behavioral discipline indicators.
7. Quiz & Micro-Test Metrics Immediate post-lecture retention scores providing rapid feedback loops on immediate concept absorption.	8. Academic Participation Doubt-asking frequency, forum activity, and classroom engagement metrics quantifying interactive learning effort.
9. Internal Assessment Marks Mid-term sessional test stability and progress consistency indicators for exam readiness.	10. Recent Trajectory Short-term academic momentum velocity, weighting recent assessments higher to capture rapid swings.

Key Architecture Takeaway: By combining behavioral signals (Attendance, Submissions, Participation) with performance metrics (Units, Labs, Quizzes), Edu Growth identifies *why* a student is lagging—distinguishing conceptual difficulty from engagement fatigue.

3. CORE INNOVATIONS & MATHEMATICAL FORMULATIONS

🌟 Dynamic Unit-Wise Mentorship Engine

Unlike legacy systems with static, semester-long mentor allocations, Edu Growth introduces dynamic, algorithmic mentorship routing. When a student or batch exhibits conceptual weakness in a specific syllabus unit (e.g., Unit 3), the system dynamically routes them to the faculty member demonstrated to have the highest pedagogical efficacy on that specific unit.

TEACHER EFFICACY SCORE (TES) MATHEMATICAL MODEL

$$TES_{u,t} = w_1 \times \text{Bigl; } (S_{avg,st2,u} - S_{avg,st1,u}) / S_{avg,st1,u} \text{ \&Bigr; } + w_2 \times M_{topic,u} + w_3 \times A_{retention,u}$$

Variable / Parameter	Definition & Operational Mechanics
$TES_{u,t}$	Quantified Teacher Efficacy Score for a specific Syllabus Unit u and Teacher t .
$S_{avg,st1,u}$, $S_{avg,st2,u}$	Mean class performance in Sessional Test 1 and Sessional Test 2 for Unit u , measuring learning growth slope.
$M_{topic,u}$	Granular topic mastery index derived from post-unit micro-assessments and quiz performance.
$A_{retention,u}$	Student lecture attendance retention coefficient specifically logged during the teaching duration of Unit u .
w_1, w_2, w_3	Calibrated weighting coefficients (where $\sum w_i = 1.0$) prioritizing gain velocity, absolute mastery, and retention.

4. ROLE-BASED FEATURE MATRIX

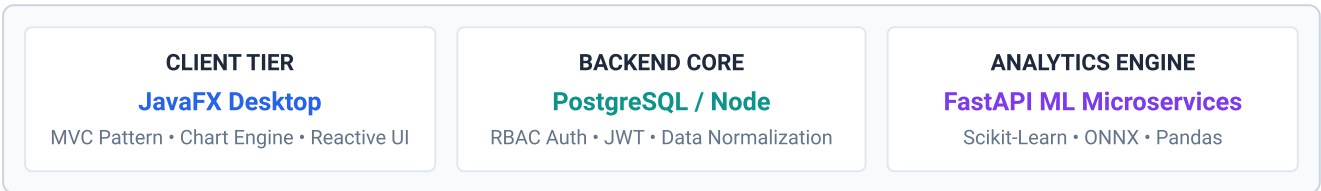
Predictive Risk & Countdown Alerts	HODTEACHER
Monitors overall risk scores and triggers early warning countdowns (e.g., "14 Days to Mid-Term – High Risk") prompting timely faculty intervention before official examinations.	
Forward Performance Prediction	STUDENTTEACHER
Utilizes Ridge/Lasso machine learning regression models on cumulative indicators to project final end-of-semester CGPA and unit scores in real time.	
Automated Root Cause Analysis (RCA)	TEACHER
Diagnoses underlying causes of performance slumps—distinguishing between missed assignment discipline issues and core conceptual understanding deficits.	
Behavioral Outlier Detection	HODTEACHER
Applies Isolation Forest unsupervised anomaly detection to highlight sudden, uncharacteristic drops in student attendance or academic engagement.	
Learning Velocity Graphs	TEACHER
Computes and plots unit-to-unit gradient improvement slopes ($\Delta \text{Mastery} / \Delta \text{Time}$) to display student academic momentum over time.	

5. TEAM RESPONSIBILITY & FEATURE OWNERSHIP MATRIX

Team **Ctrl Freaks** comprises 7 specialized engineers spanning Machine Learning, Backend Services, Desktop Frontend UI, and System Architecture:

Member	Domain	Core Feature Ownership	Technical Responsibilities
Pranav Prajapati	ML & FastAPI	- Mentor Assignment (T S) - Adaptive Assignment Allocation - Improvement Warning (T)	Engineers Teacher Efficacy Score (TES) calculation pipelines, clustering algorithms, and FastAPI endpoints for dynamic mentor allocation.
Suhani Agarwal	ML & FastAPI	- CGPA Predict (Last Data) (T S) - Risk Student Countdown (T H) - Behavioral Outlier Detection	Trains ML regression models for CGPA forecasting and Isolation Forest models for early risk anomaly detection.
Syed Rafiuddin Altamash	ML & FastAPI	- ST/PUT Unit Understanding (T S) - Concept-Wise Analysis (T S) - Learning Velocity Graphs (T)	Processes sessional test structures, extracts unit-level weak points, and computes gradient velocity metrics for charting.
Akash Raghuvanshi	ML & FastAPI	- Subject-Wise Balance (HOD) - Class/Faculty Averages (H T) - Holistic Performance Tracking	Implements macro statistical aggregations across the 10 pillars and structures top-level analytical payloads for HOD feeds.
Vivek Soni	Core Backend	- PostgreSQL Schema Design - User Auth & RBAC Security - ML Microservice Bridge	Designs normalized relational database schemas, handles JWT role authentication (Student/Teacher/HOD), and bridges ML services.
Raunak Agrahari	JavaFX UI	- Student Dashboard View - Teacher Dashboard View - HOD Analytics Panel	Develops native desktop UI components in JavaFX, integrates API data feeds, and renders live charts and risk countdowns.
Prashant Singh	UI/UX Design	- System Architecture Flow - High-Fidelity Wireframes - User Journey Mapping	Creates Figma prototypes, designs color system/asset libraries, and maps seamless user interaction pathways.

6. ECOSYSTEM SYSTEM ARCHITECTURE



7. 5-DAY SPRINT DEVELOPMENT ROADMAP

DAY 1: System Initialization, Data Structuring & Architecture Setup

OCTOBER 5

- **ML & FastAPI:** Pranav formulates TES mathematical model; Suhani cleanses historical CGPA/attendance datasets; Syed maps sessional test structures to DataFrames; Akash sets pillar weighting formulas.
- **Backend:** Vivek executes SQL scripts creating core tables (*Students, Teachers, Units, Assessments*).
- **Frontend/Design:** Prashant finalizes complete system architecture flows and initial wireframes.

DAY 2: Model Training, RBAC Security & UI Framework Setup

OCTOBER 6

- **ML & FastAPI:** Pranav trains clustering for mentor routing; Suhani fits CGPA regression models; Syed builds weak-unit extraction logic; Akash writes NumPy aggregations for class/faculty averages.
- **Backend:** Vivek develops Role-Based Access Control (RBAC) authentication routes for Student, Teacher, and HOD roles.
- **Frontend/Design:** Prashant delivers Figma UI assets; Raunak sets up JavaFX MVC structure and Login view.

DAY 3: Endpoint Wrapping, Model Export & Dashboard Construction

OCTOBER 7

- **ML & FastAPI:** Pranav builds adaptive assignment routes; Suhani exports Isolation Forest models (.pkl/.onnx); Syed exposes GET /velocity-graph; Akash finishes HOD subject-wise balance logic.
- **Backend:** Vivek configures inter-service communication pathways linking JavaFX to FastAPI ports.
- **Frontend:** Raunak completes Student Dashboard (CGPA forecast) and Teacher Dashboard (Risk alerts, velocity graphs).

DAY 4: Edge Case Hardening, Middleware & HOD Panel Finalization

OCTOBER 8

- **ML & FastAPI:** ML team handles missing historical records & unassigned mentor edge cases; syncs JSON outputs with database.
- **Backend:** Vivek implements global error handling, CORS security middleware, and input validation filters.
- **Frontend:** Raunak finishes HOD Dashboard featuring subject balance matrix and faculty performance benchmarking.

DAY 5: Full Ecosystem Integration, Dry-Run & Mentor Evaluation

OCTOBER 9

- **Morning (API Binding & Validation):** Execute Postman regression testing across all FastAPI endpoints; Raunak binds API responses directly to JavaFX chart UI elements.
- **Afternoon (Ecosystem Dry-Run):** Simulate student marks entry → trigger ML risk pipeline → verify real-time countdown alerts on Teacher & HOD dashboards.
- **Evening (Presentation):** Compile final project documentation and present the functional end-to-end platform to the Project Mentor.